Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/04/2024
PI Name:	Settles, Andrew Mark Ph.D.		
Project Title:	Understanding Genome-Wide Mutation Load in Spaceflight Culture of Cyanobacteria		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	NASA CENTER	**Phone:**	352-283-2767
Organization Name:	NASA Ames Research Center		
PI Address 1:	AST Life Support Studies		
PI Address 2:	NASA Ames Research Center		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	10/01/2021	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per F. Hernandez/ARC (Ed., 9/19/24). NOTE: End date changed to 09/30/2024 per F. Hernandez/ARC (Ed., 10/10/23). NOTE: Project dates for this investigation have changed; the period of performance (POP) is now 10/1/2021-09/30/2023 per F. Hernandez/ARC. Original POP was 09/1/2021 - 08/31/2024 (Ed. 1/19/22).		
Key Personnel Changes/Previous PI:	Nothing to report		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Increased radiation and the microgravity environment of spaceflight is expected to increase genetic mutation of all organisms. Biological Life Support systems incorporate organisms that can fix carbon dioxide and nitrogenous wastes into sugars, proteins, and other carbohydrates to release oxygen and provide feedstocks for food, fuel, or manufacturing. Plants and microbes can be used for biological support, and it is essential to understand how spaceflight impacts mutation rates in these organisms to enable appropriate countermeasures to ensure high life support productivity during long duration and deep space missions. This project will focus on the cyanobacterium Arthrospira platensis. Commonly known as Spirulina, this is an edible microalgae that is a candidate for biological life support. The project will grow A. platensis in serial cultures to allow the organism to evolve in spaceflight on the International Space Station. Whole genome sequencing of culture samples will determine the mutation load of the cyanobacteria at multiple time points throughout the long-term growth experiment. Multi-omics analysis including RNA sequencing, proteomics, metabolomics, and analytical chemistry of nutritional composition will give insights on the biological responses of the cells to spaceflight. The project directly addresses research topic 1 (Microbiology Studies in Support of Space Exploration) by studying a microorganism that is a candidate to support long-duration space exploration. The experiments proposed will reveal whether long-term growth of A. platensis will be simple to implement or if there is risk of undesirable cellular responses that require mitigation. If spaceflight is a significant stress for this species, the project will identify specific genes and pathways that would need to be targeted for genetic improvement. The project is expected to further our knowledge of microorganism adaptation to long-term growth in spaceflight and contribute towards the development of bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project primarily investigates DNA damage risks associated with low Earth orbit. The focus is on an algae crop that is grown as a protein source in human and animal diets. The fundamental knowledge developed in the project will contribute to a better understanding of DNA damage mechanisms and may reveal new targets for protecting humans or agriculture from environmental stresses.
Task Progress:	We continued with flight definition experiments to refine our science requirements and concept of operations for the flight experiment. A majority of our key research results was published in a peer-reviewed paper published by the International Conference on Environmental Systems (Fisher et al, 2024). This open access paper is available through the Texas Tech University Libraries at https:// . [Ed. Note: The paper is also available for download from the Bibliography for this NASA Task Book project. See Bibliography for Downloadable Conference Materials.] The conference paper reports pre-flight experimental results on culture conditions using flurorinated ethylene propylene (FEP) bioreactor bags, culture pH as a function of cell growth, use of glucose as a carbon source to allow dark storage of Arthrospira platensis (A. platensis) cultures, 9-month shelf-life of lyophilized and liquid media, optimal biomass dewatering method, and estimated crew time required to complete sample processing for each passage. The media shelf-life storage methods reported in Fisher et al. (2024) were not optimal for implementation in flight. Lyophilized media is very stable but requires additional crew time to rehydrate the media. Liquid media stored in FEP bioreactor bags accumulated gas within the bag over time. Although the gas did not impact A. platenesis growth, it indicated that the media composition is altered and poses a risk of over pressurizing the FEP bioreactor bags. We completed a second long-term media storage test with liquid media in FEP bioreactor bags that are placed in a Mylar secondary containment bag. This storage experiment showed 9-month old media supported A. platensis growth at a nearly identical rate to freshly prepared media. No excess gas was observed during this test, indicating that 9-month ambient storage of liquid media is acceptable to meet our science requirements. We also made significant progress in developing improved cryopreservation methods to store live A. platensis cells. Our new method has been validated for storage periods longer than 8 months and we adapted the concept of operations for flight to use cryopreserved cells for the initial inoculum of each long-term culture. We engaged the Washington State University J.J.Murdock Metabolomics Laboratory Service Center to characterize untargeted metabolomics sample analysis for A. platensis. Our initial experiment compared flask grown cultures to FEP Bioreactor bag cultures. Untargeted metabolomics of the biomass from these cultures showed eight outlier metabolites that are relatively increased in the FEP bioreactor.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Fisher KE, Broddrick JT, Settles AM. "Space Algae-2: Preflight testing for a long-duration, multi-omics analysis of Arthrospira platensis." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Conference Materials (Downloadable)	Fisher KE, Saban KE, Broddrick JT, Settles AM. "Space Algae-2 ground and lunar analog studies in preparation for long-duration propagation of cyanobacteria in spaceflight." 53rd International Conference on Environmental Systems (ICES), Louisville, Kentucky, July 21-25, 2024. , Jul-2024 ICES 2024_SA2_071524_Final.pdf
Papers from Meeting Proceedings	Fisher KE, Saban KE, Broddrick JT, Settles AM. "Space Algae-2 ground and lunar analog studies in preparation for long-duration propagation of cyanobacteria in spaceflight." 53rd International Conference on Environmental Systems (ICES), Louisville, Kentucky, July 21-25, 2024. 53rd International Conference on Environmental Systems (ICES), Louisville, Kentucky, July 21-25, 2024. ICES paper ICES-2024-184. https://hdl.handle.net/2346/98877 , Jul-2024